shifts are going to be a serious problem unless we're dealing with a really good TN panel.

In short, choose TN panels only if you value 3D and gaming over colour accuracy. These are definitely not the kind of panels you choose for colour accurate web designing, video production, and photo manipulation purposes.

IPS Panels

If TN is at one end of the LCD quality spectrum, IPS (In Plane Switching) panels lie at the very extreme. Nothing beats this technology when it comes to absolute picture quality. It delivers the trifecta of image quality, colour accuracy, and viewing angles, thereby making it indispensable for colour accurate work such as web/graphics design and photo/video editing. As expected, these monitors also happen to be at the higher end of the price spectrum. But for the money, you get sharp, crisp picture with vibrant colour reproduction and viewing angles as wide as 178 degrees. In fact, IPS panels are the way to go if you're looking for a larger display meant to accommodate more than one viewer.

Pricey IPS panels may offer much higher image fidelity than TN panels, but the technology still has its Achilles heel. The response time of this technology is the slowest of the lot, and ranges from 6 ms to 16 ms depending upon how deep your pockets can go. Make no mistake, this is only a tad bit slower than the average TN panel and not even noticeable in day to day usage. However, slower response time can be telling in video games. If you're an FPS gamer who's particular about image quality, you might want to choose IPS panels that deliver a response time of 8 ms or lower.

Dig deeper within the IPS hierarchy and you'll find further subdivisions that are differentiated by their inherent pixel structure. Each variant has its own pros and cons, wherein parameters such as colour accuracy, contrast, and black levels vary according to the technology employed. S-IPS panels, for example, cost a premium and can be identified by the presence of a purple hue on blacks, when the screen is viewed from chronically acute angles. The cost factor makes this IPS type not as widespread. This is especially true of the more modern H-IPS panels, which incorporate a pixel structure that leads to improved contrast ratios and finer pixel pitch for improved image quality.

Most affordable IPS panels, however, are e-IPS that are pretty decent image quality wise, but only offer 6-bit colour depth. This is rarely an issue considering modern video processing hardware does a pretty good job at FRC/dithering to achieve true colour output despite the 6-bit panel limitation. H-IPS and S-IPS panels with true 8-bit panels are usually necessary in prohibitively expensive colour accurate monitors for professionals.

VA Panels

Vertical Alignment (VA) panels lie somewhere in the middle of the cost/quality spectrum of LCD displays. This panel type is further dubdivided into S-PVA/MVA and offer significantly better colour as well as generous viewing angles when compared to TN panels. Again, like the IPS panels, VA panels tend to have slower response times. Since this technology is more or less similar to S-IPS, it also faces the same problems. Even though VA panels deliver great colour performance and wide viewing angles, it's still not as good as IPS panels. To make matters worse, the response times is worse when compared to IPS. These panels are definitely not the right fit for gamers, because they tend to add palpable input lag as well. On the flip side, VA panels tend to offer better blacks even when compared to IPS due to their inherently better contrast ratios. Unfortunately, the performance isn't as balanced because these panels are marred by serious colour shift issues, which also lead to annoying brightness levels across the width, especially when viewing even from a slight off centre angle. What's more, even the central viewing position isn't immune to issues. Colour shifts tend to cause reduced darker details

in dark scenes, even when viewed directly from the centre. While some users aren't sensitive to these colour shifts and uneven brightness, but still others deem it a deal-breaking issue. Thankfully the proliferation of cost effective IPS panels means that VA technology has more or less been edged out. You're better off avoiding monitors with these panels, since you're likely to find a good IPS alternative at roughly the same price range.

The World Beyond LCD

LED-backlit LCDs is doing great in the computer monitor space, but once you enter the realm of larger screens, it opens up the market for OLED alternatives. The sublime plasma technology may be dead, but it's throne is taken over by another capable emissive alternative that is OLED, which fills in the gaps left behind by LED displays. What follows is an overview of how the emissive OLED displays trump transmissive LED panels in crucial picture quality parameters.

Rich Colours

Colours in LCD panels are limited by their rather convoluted means of colour reproduction. These displays essentially use liquid-crystal filters to block and bend light to create different colours. This isn't the most fool-proof or efficient means of creating colours, and is marred further by optical anomalies that compromise colour fidelity. On